# Bloomberg Sketchbook

## Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

## How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

# Your reading path

1. Bloomberg: passing the four tests
2. Economic indicators
3. Currencies
4. Fixed income and the yield curve
5. Equities and valuation multiples

Then: paper labs where available, lesson-check answers, and subject practice.

# Lesson 1: Bloomberg: passing the four tests

The Bloomberg course is pass/fail on four tests: at least 70% on each one, done by December 14, under class code FM3NH456BD.

## Start with an everyday idea

Bloomberg is a certification task: learn the required modules, take the official tests on Bloomberg, and verify your recorded results. This app only helps you prepare.

## Words you need

BMC means Bloomberg Market Concepts. Your supplied requirements say at least 70% on each of four required tests. An app practice answer is not an official test result.

## Build the explanation

Sign up with the access guide in your folder and enter class code FM3NH456BD so the instructor can see your scores.

Watching videos or getting a certificate is not enough. All four test scores must be 70% or more.

Do the lessons actively: pause, note unfamiliar terms, and link them to what you learned here (yields and duration, futures, options, credit).

Aim to finish by your own target of December 6, leaving a week of buffer before the real deadline of December 14.

After each test, check the score shows on your class record.

## Formula and units

Pass = all four tests ≥ 70%

Deadline: Mon Dec 14, 2026 (target Dec 6)

Bloomberg lessons and tests are online, not in your folder.

## Walk through the sketch

1. Step 1: sign up and enter class code FM3NH456BD.
2. Then four tests along the way. Each needs at least 70%.
3. Your own target finish line is December 6.
4. The real deadline is December 14. The week in between is your safety buffer.

## Worked example

Plan: you have 30-minute Bloomberg slots twice a week from Nov 2. How do you fit four tests before Dec 6?

1. Week of Nov 2: register, confirm the class code, start the first module.
2. Weeks of Nov 9 and Nov 16: finish modules and take tests 1 and 2.
3. Weeks of Nov 23 and Nov 30: tests 3 and 4 (book a longer session if a test needs it).
4. Retake any test under 70%, then check all four scores appear.

Four tests done by Dec 6, one week of buffer

## Common mistakes

1. Not entering the class code, so the instructor never sees your scores.
2. Assuming a certificate means you passed.
3. Leaving all four tests to the final week.

## Try before checking

One test is 65% but your average is 80%. Have you met the supplied requirement?

1. Yes, the average is enough
2. No, each required test must reach 70%
3. Yes, if the videos are complete

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE5500_NOTES_01_Bloomberg_Signup_and_BMC_Access_Summer_2025.pdf; FRE5500_NOTES_02_Requirements.md

# Lesson 2: Economic indicators

Markets move on surprises, not on the news itself: what matters is the actual number minus what economists expected.

## Start with an everyday idea

If everyone expected 180,000 new jobs but the release says 250,000, the surprise is +70,000. Markets react to surprises and details, not just whether a headline sounds good.

## Words you need

GDP measures economic output. Inflation is price growth. Consensus is the expected figure. An indicator is a measured clue about the economy.

## Build the explanation

GDP: total output. Growth rate is the headline. Released quarterly.

Inflation (CPI, PCE): how fast prices rise. Central banks target about 2%.

Jobs (nonfarm payrolls, unemployment rate): monthly; one of the biggest market movers in the US.

Surveys (PMI, consumer confidence): early signals. PMI above 50 = expanding.

Leading vs lagging: leading indicators (PMI, building permits) turn before the economy; lagging ones (unemployment) after.

Central bank link: strong growth or high inflation → expect higher policy rates → bond yields up, currency up, often stocks down.

On the terminal, the economic calendar shows the release time, the consensus (survey) and the actual.

## Formula and units

Surprise = Actual - Consensus

Positive surprise (strong data) → yields ↑, currency ↑

Negative surprise → yields ↓, currency ↓

Typical reactions. Real markets can react differently if the details disagree with the headline.

## Walk through the sketch

1. The economists' consensus forecast: +180k jobs.

2. The release: +250k. A +70k surprise.

3. The market reacts within seconds: 2-year yield jumps up.

4. And the dollar rises against other currencies.

## Worked example

Consensus for US payrolls is +180k. The actual is +250k. Unemployment is unchanged. What do you expect for Treasury yields and the dollar?

1. Surprise = 250k - 180k = +70k, a strong beat

2. Stronger jobs → the Fed is less likely to cut rates

3. Short-term Treasury yields rise (prices fall)

4. The dollar strengthens, as higher US rates attract money

Yields up, dollar up

## Common mistakes

1. Reacting to the level instead of the surprise.

2. Mixing up month-over-month and year-over-year inflation.

3. Forgetting that a bond price falls when its yield rises.

## Try before checking

Expected jobs 180k, actual 250k. Surprise?

1. +70k

2. +250k

3. -70k

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE5500_NOTES_02_Requirements.md; Bloomberg Market Concepts: Economic Indicators module

# Lesson 3: Currencies

An exchange rate is just the price of one currency in another. Interest rates, inflation, growth and risk appetite push it around.

## Start with an everyday idea

A currency quote is a price tag with two currency units. EUR/USD 1.10 says one euro costs 1.10 US dollars. Write the units before multiplying.

## Words you need

Base currency is the first currency in the pair. Quote currency is the second. Appreciation means gaining value relative to the other currency.

## Build the explanation

Quote: EUR/USD = 1.10 means 1 EUR (the base) = 1.10 USD (the quote currency). If it rises to 1.12, the euro got stronger.

Pips: the 4th decimal (0.0001). Spread = ask - bid.

Interest rate differentials: higher rates attract money → currency tends to strengthen (spot). Forwards follow interest rate parity (see the Risk class on currencies).

Carry trade: borrow in a low-rate currency, invest in a high-rate one. Earns the gap, until a sudden reversal.

PPP (purchasing power parity): in the long run, the same basket should cost the same everywhere; high-inflation currencies weaken.

Safe havens (USD, JPY, CHF) tend to rise in a crisis.

Who cares: exporters, importers, investors holding foreign assets, central banks.

## Formula and units

Amount in quote currency = amount in base × rate

Cross rate: EUR/JPY = EUR/USD × USD/JPY

% change of base = new rate / old rate - 1

Always write the pair as BASE/QUOTE and keep units: QUOTE per 1 BASE.

## Walk through the sketch

1. A seesaw: EUR on the left, USD on the right. At 1.10, one euro balances 1.10 dollars.

2. The ECB raises rates; money flows into euros. The euro side gets heavier: EUR/USD rises to 1.155.

3. Cross rates: if you know EUR/USD and USD/JPY you can get EUR/JPY by multiplying.

## Worked example

EUR/USD = 1.10 and USD/JPY = 150. (a) How many dollars is €2m? (b) What is EUR/JPY? (c) EUR/USD moves to 1.155. How much did the euro gain?

1. (a) 2,000,000 × 1.10 = $2,200,000

2. (b) EUR/JPY = 1.10 × 150 = 165 yen per euro

3. (c) 1.155 / 1.10 - 1 = +5%

4. So a US investor holding euro assets gained 5% from the currency alone.

$2.2m; EUR/JPY 165; euro +5%

## Common mistakes

1. Multiplying when you should divide (check units: USD per EUR × EUR = USD).

2. Reading a fall in EUR/USD as the dollar weakening (it is the dollar strengthening).

3. Forgetting the currency effect on foreign investments.

## Try before checking

At EUR/USD 1.10, €20 buys how many dollars?

1. $18.18

2. $22

3. $20

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Currencies module

# Lesson 4: Fixed income and the yield curve

A bond's price and yield move in opposite directions. The yield curve (yields by maturity) shows what the market expects for rates and growth.

## Start with an everyday idea

A yield curve compares borrowing rates for different lengths of time. A credit spread compares a riskier borrower’s yield with a suitable reference yield.

## Words you need

Maturity is time until repayment. A basis point is 0.01 percentage point: 100 bp = 1 percentage point. Inverted means shorter-term yields exceed longer-term yields.

Helpful earlier topics: Bonds and building a zero curve, Forward rates, duration and convexity

## Build the explanation

Price vs yield: rates up → existing bonds with lower coupons are worth less → price down.

Yield curve shapes: normal (upward: longer = higher yield), flat, inverted (short rates above long, often seen before recessions).

Central banks set short rates; long rates reflect expected future short rates plus a term premium.

Credit spread: corporate bond yield - Treasury yield of the same maturity. Wider spread = more perceived default risk.

Ratings: AAA ... BBB- is investment grade; BB+ and below is high yield ("junk").

Duration (see the Derivatives class) tells you price sensitivity: duration 7 → about -7% for +1% in yield.

## Formula and units

Credit spread = corporate yield - Treasury yield
Curve slope = 10-yr yield - 2-yr yield
% price change ≈ -Duration × Δyield

1 basis point (bp) = 0.01%.

## Walk through the sketch

1. A normal yield curve: you get paid more to lend for longer.

2. A flat curve: all maturities pay about the same.

3. An inverted curve: short rates above long rates. Markets expect cuts, often a recession warning.

4. Add a company's curve above the Treasury curve. The gap is the credit spread: 130 bp at 10 years.

## Worked example

10-year Treasury yields 4.20%. A BBB company's 10-year bond yields 5.50%. The 2-year Treasury yields 4.60%. Spread? Curve shape? If the corporate bond (duration 8) sees its yield rise 0.25%, price change?

1. Credit spread = 5.50 - 4.20 = 1.30% = 130 bp

2. Slope = 4.20 - 4.60 = -40 bp → the curve is inverted

3. Price change ≈ -8 × 0.25% = -2%

4. Inverted curve: markets expect rate cuts ahead, often linked with slowing growth.

Spread 130 bp; inverted curve; price -2%

## Common mistakes

1. Saying a bond price rises when yields rise.

2. Comparing yields of different maturities when computing a spread.

3. Writing 130 bp as 13%.

## Try before checking

Corporate yield 5.5%, Treasury yield 4.2%. Spread?

1. 13 bp

2. 130 bp

3. 1,300 bp

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Fixed Income module; FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf

# Lesson 5: Equities and valuation multiples

A share is a slice of a company's future profits. Multiples like P/E are a quick way to compare how "expensive" companies are.

## Start with an everyday idea

A share is a small ownership stake. If a share costs $50 and annual earnings per share are $2.50, you pay 20 times that annual earnings amount. That is a comparison, not a guaranteed payback time.

## Words you need

EPS is earnings per share. P/E is price divided by EPS. A dividend is a cash distribution. Enterprise value includes debt and subtracts cash in the simplified formula.

Helpful earlier topics: Money today vs money later, Portfolio theory and the CAPM

## Build the explanation

P/E = price / earnings per share. How many dollars you pay for $1 of yearly profit. Higher can mean faster expected growth, or overpricing.

EV/EBITDA: enterprise value (equity + debt - cash) over operating earnings. Compares firms with different debt levels.

Dividend yield = dividend / price. Market cap = price × shares.

DCF (discounted cash flow): the "proper" value: PV of future free cash flows (Valuation classes). A simple version is the Gordon model: P = D1/(r - g).

Indices (S&P 500, etc.) track groups of stocks. Beta measures how much a stock moves with the index (Portfolio/CAPM class).

What moves stocks: earnings surprises, guidance, interest rates (higher rates = lower PV of future profits), sector news.

## Formula and units

P/E = Price / EPS

EV = Market cap + Debt - Cash

Gordon growth: P0 = D1 / (r - g)

EPS = earnings per share, D1 = next year's dividend, r = required return, g = growth rate.

## Walk through the sketch

1. A $50 share. Each year it earns $2.50 per share.
2. So you pay 20 years of today's profit: P/E = 20×.
3. The Gordon model values the growing stream of dividends: 1.50 / (9% - 6%) = $50.
4. Small changes in growth move the value a lot: at 5% growth the value drops to $37.50.

## Worked example

A stock is $50 with EPS $2.50 and next year's dividend $1.50. Required return 9%, dividends grow 6% a year. P/E? Gordon value? Is it cheap?

1. P/E = 50 / 2.50 = 20×
2. Gordon value = 1.50 / (0.09 - 0.06) = 1.50 / 0.03 = $50
3. Model value = market price, so fairly priced under these assumptions.
4. If growth were only 5%: 1.50 / 0.04 = $37.50, so the price assumes 6% growth.

P/E 20×; Gordon value $50 (fairly priced)

## Common mistakes

1. Comparing P/E across very different industries.
2. Using Gordon when g ≥ r (the formula breaks).
3. Forgetting to subtract cash and add debt for EV.

## Try before checking

Price $50, EPS $2.50. P/E?

1. 20×
2. 2×
3. 125×

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Equities module; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf

# Lesson-check answers

## 1. Bloomberg: passing the four tests

Answer: No, each required test must reach 70%. The supplied October 30 requirements specify at least 70% on each of four tests, not a 70% average.

## 2. Economic indicators

Answer: +70k. Actual minus expected = 250k - 180k = +70k. The market response is not guaranteed.

## 3. Currencies

Answer: $22. 20 EUR × 1.10 USD per EUR = 22 USD. The EUR units cancel.

## 4. Fixed income and the yield curve

Answer: 130 bp. 5.5% - 4.2% = 1.3 percentage points = 130 basis points.

## 5. Equities and valuation multiples

Answer: 20×. 50 ÷ 2.50 = 20. A multiple by itself does not tell you whether a stock is a good investment.

# Bloomberg: original warm-up practice

These 32 questions and four cases are original learning aids. They were not copied from real BMC tests. No official BMC assessment questions were supplied. This practice neither counts toward certification nor predicts your official score.

Complete the assigned modules and assessments in the Bloomberg for Education portal. Follow the requirements supplied for your NYU class, and verify any changes with your instructor.

# Economic Indicators - try first

### 1. What does GDP (gross domestic product) measure?

A. The total value of goods and services a country produces in a period B. The total value of a country’s stock market C. How much the government spends each year

### 2. Economists expected 200,000 new jobs. The report shows 150,000. What is the surprise?

A. +150,000 B. -50,000 C. +50,000

### 3. What does the CPI (consumer price index) track?

A. Company profits B. Changes in the price of a typical basket of things people buy C. Interest rates on government bonds

### 4. What is the difference between real and nominal GDP?

A. Real GDP removes the effect of price changes (inflation) B. Real GDP only counts physical goods, not services C. There is no difference

### 5. Inflation is running well above the central bank’s target. What is the central bank most likely to do?

A. Cut interest rates B. Raise interest rates C. Print a new currency

### 6. A manufacturing PMI reading of 55 usually suggests…

A. Manufacturing is shrinking B. Manufacturing is growing C. Prices are falling

### 7. Which indicator is usually described as “lagging” (it changes after the economy has already turned)?

A. The unemployment rate B. Stock market prices C. New building permits

### 8. Why can a “good” economic number make a market fall?

A. Markets only look at bad news B. The number was still worse than what traders expected, or it raises the chance of higher interest rates C. Economic data never affects markets

# Economic Indicators - explanations

1. The total value of goods and services a country produces in a period. GDP adds up the value of everything produced in the economy over a quarter or a year. It is the broadest measure of economic output.

2. -50,000. Surprise = actual - expected = 150,000 - 200,000 = -50,000. Markets react to the surprise, not just the headline number.

3. Changes in the price of a typical basket of things people buy. CPI follows the cost of a fixed basket of consumer goods and services. Its yearly percentage change is a common measure of inflation.

4. Real GDP removes the effect of price changes (inflation). Nominal GDP uses today’s prices. Real GDP adjusts for inflation, so it shows whether the economy actually produced more, not just charged more.

5. Raise interest rates. Higher policy rates make borrowing more expensive, which slows spending and helps bring inflation down.

6. Manufacturing is growing. A purchasing managers’ index above 50 signals expansion; below 50 signals contraction.

7. The unemployment rate. Firms usually hire and fire after demand has already changed, so unemployment tends to confirm a turn rather than predict it.

8. The number was still worse than what traders expected, or it raises the chance of higher interest rates. Prices already reflect expectations. A strong number can disappoint if expectations were higher, or worry investors about rate hikes.

# Currencies - try first

### 1. EUR/USD = 1.10 means...

A. 1 US dollar buys 1.10 euros B. 1 euro costs 1.10 US dollars C. The euro is worth 10% less than the dollar

### 2. EUR/USD moves from 1.10 to 1.15. What happened?

A. The euro strengthened against the dollar B. The euro weakened against the dollar C. Nothing; quotes do not show strength

### 3. USD/JPY = 150. How many yen do you get for $1,000?

A. 6.67 yen B. 150 yen C. 150,000 yen

### 4. All else equal, if a country raises its interest rates, its currency tends to...

A. Strengthen, because investors want the higher return B. Weaken, because borrowing costs rise C. Stay exactly the same

### 5. The home currency weakens. Who usually benefits most?

A. People importing foreign goods B. Exporters selling abroad C. Tourists travelling overseas

### 6. Country A has a higher interest rate than country B. In the forward market, A’s currency usually trades...

A. At a forward discount (cheaper for future delivery) B. At a forward premium (more expensive for future delivery) C. At exactly the spot rate

### 7. During a market panic, which currencies have often been treated as “safe havens”?

A. Japanese yen and Swiss franc B. Currencies of fast-growing emerging markets C. None; all currencies fall together

### 8. A country imports much more than it exports for many years. All else equal, this tends to put what pressure on its currency?

A. Upward pressure B. Downward pressure C. No pressure

# Currencies - explanations

1. 1 euro costs 1.10 US dollars. The first currency (EUR) is the base. The quote tells you how many units of the second currency (USD) one euro costs.

2. The euro strengthened against the dollar. One euro now buys more dollars (1.15 instead of 1.10), so the euro gained value relative to the dollar.

3. 150,000 yen. Each dollar buys 150 yen, so $1,000 × 150 = 150,000 yen.

4. Strengthen, because investors want the higher return. Higher rates attract foreign money looking for better returns. Buying that currency pushes its value up.

5. Exporters selling abroad. A weaker home currency makes exported goods cheaper for foreign buyers, which helps exporters sell more.

6. At a forward discount (cheaper for future delivery). Covered interest parity: the extra interest you earn in A is offset by buying A’s currency back more cheaply in the forward contract.

7. Japanese yen and Swiss franc. Investors under stress have historically moved money into currencies such as the yen, the Swiss franc and often the US dollar.

8. Downward pressure. To pay for imports, the country sells its own currency for foreign currency. Persistent net selling tends to weaken it over time, though capital flows can offset this.

# Fixed Income - try first

1. Market interest rates (yields) rise. What happens to the price of an existing fixed-coupon bond?

A. It rises B. It falls C. It stays the same

2. A bond pays a 5% coupon. Similar bonds now yield 6%. The bond will trade…

A. Below face value (at a discount) B. At face value (par) C. Above face value (at a premium)

3. How much is one basis point?

A. 1% B. 0.1% C. 0.01%

4. The 2-year yield is above the 10-year yield. This yield curve is called…

A. Normal (upward sloping) B. Inverted C. Flat by definition

5. A company’s credit spread widens from 1.5% to 3%. Investors are signalling…

A. They see more default risk B. The company has become safer C. Government rates have fallen

6. Which bond is usually more sensitive to a change in interest rates?

A. A 2-year bond B. A 30-year bond C. Both are equally sensitive

7. A bond has a modified duration of 7. Yields rise by 1 percentage point. Roughly how does its price change?

A. Falls about 7% B. Rises about 7% C. Falls about 0.7%

8. On the S&P scale, what is the lowest rating still called “investment grade”?

A. A B. BBB- C. BB+

# Fixed Income - explanations

1. It falls. New bonds now pay more, so an old bond with a lower coupon must become cheaper to compete. Bond prices and yields move in opposite directions.

2. Below face value (at a discount). Buyers will only accept the lower 5% coupon if they pay less than face value, so the price falls below par.

3. 0.01%. A basis point is one hundredth of a percentage point: 100 bp = 1%.

4. Inverted. An inverted curve means short-term rates are higher than long-term rates. Markets often read it as a warning of a future slowdown.

5. They see more default risk. The spread is the extra yield over a safe government bond. A wider spread means investors want more compensation for credit risk.

6. A 30-year bond. Longer bonds have higher duration: more of their cash arrives far in the future, so discounting changes affect them more.

7. Falls about 7%. Price change ≈ -duration × change in yield = -7 × 1% = -7%. Convexity makes the true fall a little smaller.

8. BBB-. BBB- and above is investment grade. BB+ and below is high-yield, sometimes called “junk”.

# Equities - try first

1. A share costs $50 and earnings per share are $2.50. What is the P/E ratio?

A. 2.5 B. 20 C. 125

2. How is market capitalisation calculated?

A. Share price × number of shares outstanding B. Revenue × profit margin C. Total debt + cash

3. A $50 share pays $2 of dividends a year. What is the dividend yield?

A. 2% B. 4% C. 25%

4. A stock has a beta of 1.5. If the market rises 10%, the stock is expected to…

A. Rise about 15% B. Rise about 10% C. Fall about 15%

5. A company buys back some of its own shares. If total earnings stay the same, earnings per share…

A. Fall B. Rise C. Stay the same

6. Compared with “value” stocks, “growth” stocks usually have…

A. Lower P/E ratios B. Higher P/E ratios C. No earnings at all

7. A company beats this quarter’s earnings forecast, but the stock falls. A likely reason is…

A. It lowered its guidance for future quarters B. Stock prices ignore earnings C. The beat was too large

8. Market cap $100m, debt $30m, cash $10m. What is enterprise value (EV)?

A. $120m B. $140m C. $80m

# Equities - explanations

1. 20. P/E = price ÷ earnings per share = 50 ÷ 2.50 = 20. Investors pay $20 for every $1 of yearly earnings.

2. Share price × number of shares outstanding. Market cap is what the market says all the company’s shares are worth together.

3. 4%. Dividend yield = annual dividend ÷ price = 2 ÷ 50 = 4%.

4. Rise about 15%. Beta measures sensitivity to the market: 1.5 × 10% = about 15%. It is an average tendency, not a guarantee.

5. Rise. The same earnings are now divided among fewer shares, so EPS rises.

6. Higher P/E ratios. Investors pay more today for growth stocks because they expect earnings to rise quickly in the future.

7. It lowered its guidance for future quarters. Prices look forward. Weak guidance about future earnings can outweigh a good quarter that has already happened.

8. $120m. EV = market cap + debt - cash = 100 + 30 - 10 = $120m. It is roughly what it costs to buy the whole business.

# Market case: Read the surprise, not just the headline

Markets expected annual inflation of 3.0%. The release is 3.4%, down from 3.6% previously. Two-year government yields rise after the release.

1. Calculate the change from the previous release and the surprise versus expectations.

2. Explain how falling inflation and rising yields can occur together. What would you check before claiming causation?

## Work through the explanation

1. Inflation fell by 0.2 percentage points from the previous release, but was 0.4 percentage points above expectations. Those are different comparisons.

2. Investors may revise expected policy rates upward relative to their prior expectations, even while inflation is declining. Prices respond to new information relative to what was already priced in.

3. Check release timing, core inflation, revisions, other simultaneous news and changes in expected policy. A same-day move alone does not prove which news caused it.

# Market case: Keep the currency quote straight

A US student must pay EUR 2,000. EUR/USD rises from 1.08 to 1.12 USD per EUR. Ignore fees and bid/ask spreads.

1. Which currency strengthened? Calculate the dollar cost before and after.

2. What is the percentage rise in EUR/USD? Is the percentage change in its reciprocal exactly the negative of that number?

## Work through the explanation

1. The euro strengthened: each euro now buys more dollars. The payment costs $2,160 before and $2,240 after, an increase of $80.

2. EUR/USD rose by 1.12/1.08 - 1 = 3.7037%.

3. The reciprocal USD/EUR changed by 1.08/1.12 - 1 = -3.5714%. Reciprocal percentage returns have different denominators, so they are not exact opposites.

# Market case: Separate rate risk from credit risk

A corporate bond yields 6.0%, while a comparable government bond yields 4.0%. Later these are 6.4% and 4.6%. The corporate bond has modified duration 5.0.

1. Calculate the credit spread at each date. Did it widen or narrow?

2. Estimate the corporate bond’s percentage price change from its yield change. Explain why its price can fall even as its credit spread narrows.

## Work through the explanation

1. Initial spread = 2.0 percentage points (200 bp); later = 1.8 points (180 bp). It narrowed by 20 bp.

2. The corporate yield rose 0.4 percentage points. Approximate price change = -5 × 0.004 = -2.0%, ignoring convexity and coupon income.

3. The government yield rose 60 bp while the spread narrowed 20 bp, leaving a net corporate-yield rise of 40 bp. Narrower spreads do not guarantee rising prices.

# Market case: Read an earnings headline critically

A company earns $100m and has 50m shares. Its stock is $40. It buys back 5m shares; assume earnings and stock price stay unchanged for this exercise.

1. Calculate EPS and P/E before and after the buyback.

2. Has the business necessarily created economic value just because EPS rose? Name information you would need.

## Work through the explanation

1. Before: EPS = 100/50 = $2; P/E = 40/2 = 20.

2. After: 45m shares remain, EPS = 100/45 = $2.2222 and P/E = 40/2.2222 = 18.

3. EPS increased because the denominator shrank. Examine the repurchase price, financing, lost interest income, investment alternatives and leverage. Holding earnings and price fixed isolates the arithmetic, not the full economic effect.